

FPGA Programming – VHDL Part

VHDL by Examples – Combinatorial

Andrea Triossi – University of Padova – INFN Padova

VHDL basic structures

- **entity**: it is a black box where only the interface signals (**ports**) are described
- **architecture**: it describes the content of the box in terms of functionalities and/or structures of the circuits

```
entity And2 is  
    port (x,y: in BIT; z: out BIT);  
end entity And2;
```

```
architecture ex1 of And2 is  
begin  
    z <= x and y;  
end architecture ex1;
```

```
architecture ex2 of And2 is  
    signal w : BIT_VECTOR(1 downto 0);  
begin  
    z <= '1' when w = "11" else  
        '0';  
    w <= x & y  
end architecture ex2;
```

VHDL instantiation

- Instead of directly describe the functionality, a more structured (hierarchical) description can be given
- Sub-blocks instantiation

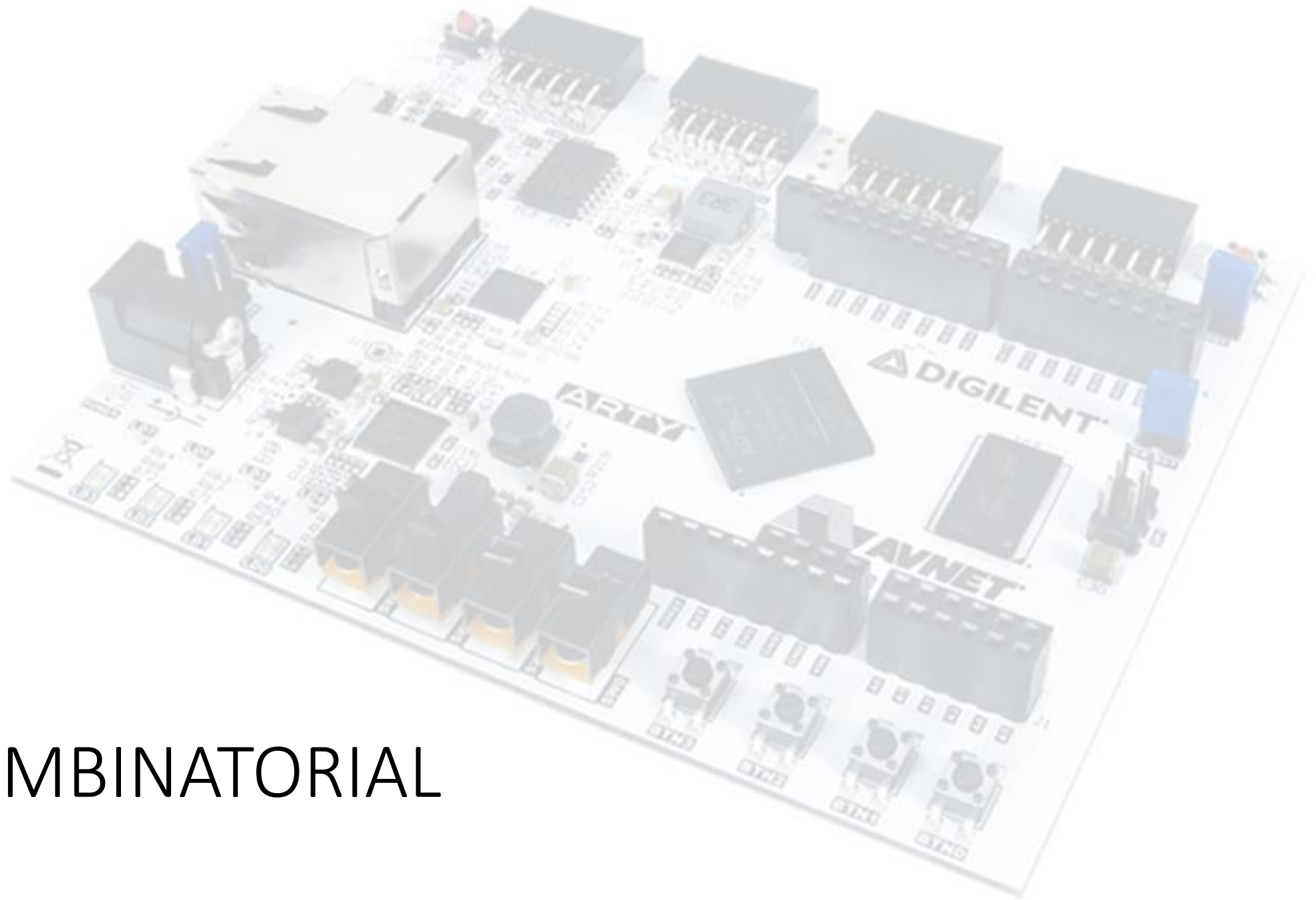
```
entity comb_function is  
  port (a, b, c : in BIT; z: out BIT);  
end entity comb_function;  
architecture expression of comb_function is  
begin  
  z <= (not a and b) or (a and c);  
end architecture expression;
```

VHDL instantiation

```
entity Or2 is  
  port (x, y : in BIT; z: out BIT);  
end entity Or2;  
architecture ex1 of Or2 is  
begin  
  z <= x or y;  
end architecture ex1;
```

```
entity Not1 is  
  port (x : in BIT; z: out BIT);  
end entity Not1;  
architecture ex1 of Not1 is  
begin  
  z <= not x;  
end architecture ex1;
```

```
entity comb_function is  
  port (a, b, c : in BIT; z: out BIT);  
end entity comb_function;  
architecture netlist of comb_function is  
  component Not1 is ...  
  component And2 is ...  
  component Or2 is ...  
  signal p, q, r : BIT;  
begin  
  g1: Not1 port map (a, p);  
  g2: And2 port map (p, b, q);  
  g3: And2 port map (a, c, r);  
  g4: Or2 port map (q, r, z);  
end architecture netlist;
```



LAB COMBINATORIAL